

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457157.8885 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.04
Transit depth (Rp/Rs)^2	1.9e-05 +/- 0.0035 [%]
Orbital Inclination (inc)	81.32 +/- 2.92
Airmass coefficient 1 (a1)	2.4 +/- 0.35
Airmass coefficient 2 (a2)	-0.86 +/- 0.24
Scatter in the residuals of the lightcurve fit is	13.76 %
Best Comparison Star	#2 - [435, 297]
Optimal Aperture	49.55
Optimal Annulus	97.75
Transit Duration (day)	0.073 +/- 0.036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.04 vs 0.10538 (deviation 2.63σ)
Duration fit vs published	0.073 d vs 0.1167 d (deviation 1.21σ)
Mid-time O–C	3.4 min
Depth SNR	0.0
Residual scatter	13.76 %

Light curve

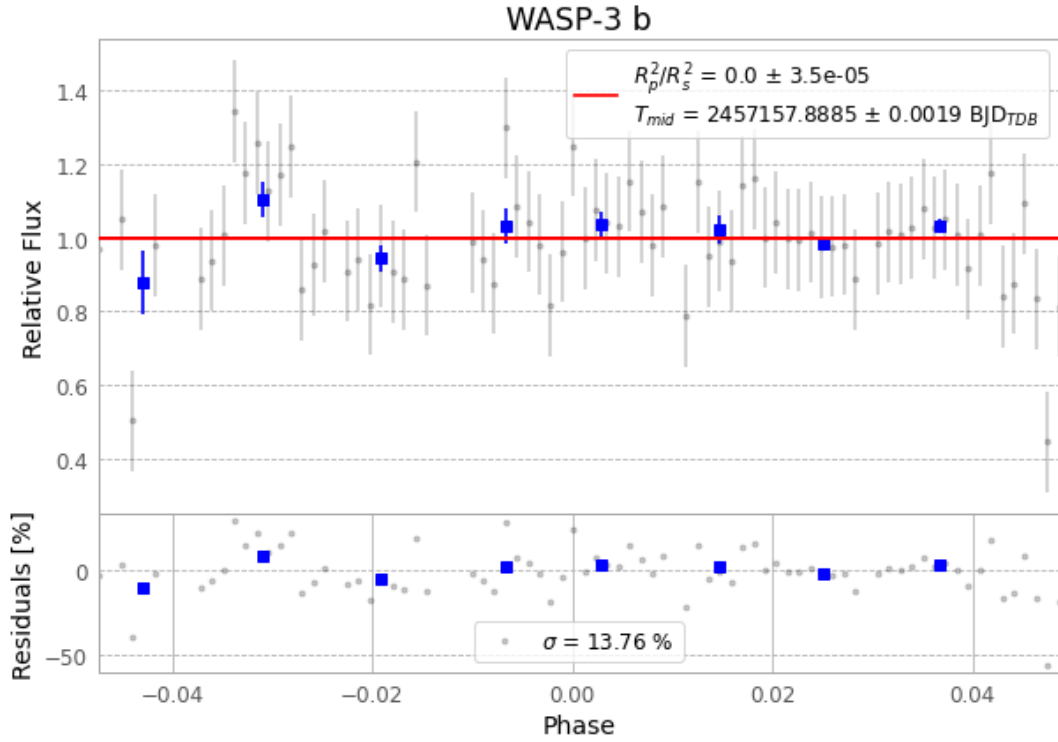


Figure 1 — FinalLightCurve_WASP-3 b_2015-05-15.png (SHA-256 584099f45255695f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [427, 254], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a29fff6529e24cbb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

86 FITS frames (56 MB), WASP-3150515070912.FITS → WASP-3150515112412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-150515.log (92ac4e4391ab...), FinalLightCurve_WASP-3 b_2015-05-15.png (584099f45255...), FinalLightCurve_WASP-3 b_2015-05-15.csv (0c2b1873ebea...)

Compute resources

Wall time 676.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 15:19Z.